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Optimization of Wear Resistance and Scratch Hardness of Boron Carbide Reinforced Polyester Powder Coatings Using Taguchi Method

Chandrakant Sawant[1] ***Ravindra Karvande***[2] & ***Vilas Jadhav***[3]

[1,2,3] *Department of Mechanical Engineering, MSS's College of Engineering and Technology, Jalna, MH, India- 431203*

Corresponding: cads.4677@gmail.com

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ABSTRACT

Polyester powder coatings are extensively used for corrosion protection and surface finishing due to their environmental friendliness, strong adhesion, and thermal stability. However, their application in tribologically demanding environments is limited by poor wear resistance and scratch hardness. In the present study, boron carbide (B_4C) nanoparticles were incorporated into polyester powder coatings to enhance mechanical and tribological performance. The coatings were deposited on mild steel substrates using electrostatic spray deposition followed by thermal curing. A systematic experimental investigation was conducted to evaluate the effects of nanofiller content, applied load, and sliding frequency on wear rate. One-Variable-At-a-Time (OVAT) analysis was initially performed to identify suitable parameter ranges, followed by Taguchi L9 orthogonal array design for multi-objective optimization. Wear testing was carried out using a reciprocating tribometer under dry sliding conditions, and scratch hardness was evaluated using a scratch tester. The results indicate that optimal B_4C reinforcement significantly improves wear resistance and scratch hardness. Statistical analysis using signal-to-noise ratio and ANOVA revealed that nanofiller content is the most influential parameter affecting wear behavior. The optimized coating system demonstrates enhanced tribological performance, making it suitable for advanced industrial applications.

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1. Introduction

Surface engineering plays a critical role in enhancing the performance, durability, and service life of engineering components operating under conditions of mechanical wear, corrosion, and aggressive environments. Among various surface protection techniques, powder coating has emerged as an environmentally sustainable and industrially preferred alternative to conventional liquid coatings due to its solvent-free processing, high material utilization efficiency, superior surface finish, and compliance with environmental regulations [1].

Powder coatings are applied as finely divided solid particles that are electrostatically deposited onto a grounded substrate and subsequently cured at elevated temperatures to form a continuous and adherent coating. The absence of volatile organic compounds (VOCs), reduced material wastage, and excellent mechanical and chemical resistance make powder coatings widely applicable in automotive, appliance, construction, electrical, and general industrial sectors [2].

Among thermosetting powder coatings, polyester-based coatings are extensively used because of their excellent corrosion resistance, thermal stability, weatherability, and aesthetic properties, particularly for outdoor applications [3-5]. However, despite these advantages, conventional polyester powder coatings exhibit limited wear resistance and scratch hardness, which restrict their application in components subjected to sliding contact, abrasion, and high mechanical stresses [6-11].

To overcome these limitations, recent research has focused on the development of polyester powder coating nanocomposites by incorporating hard ceramic and carbon-based nanofillers [12]. Among various reinforcements, boron carbide (B_4C) is particularly attractive due to its extremely high hardness, low density, excellent wear resistance, and chemical stability. The incorporation of B_4C nanoparticles into polyester matrices has the potential to significantly enhance tribological performance while retaining the inherent advantages of powder coatings [13].

In addition to material composition, the performance of polyester powder coatings is strongly influenced by process and operating parameters such as nanofiller content, applied load, sliding frequency, curing conditions, and coating thickness. Therefore, statistical optimization techniques, especially the Taguchi method and analysis of variance (ANOVA), are widely used to systematically evaluate the influence of these parameters and to identify optimal conditions with minimal experimental effort [14, 15].

The present study aims to enhance the wear resistance and scratch hardness of polyester powder coatings through boron carbide nanoparticle reinforcement and to optimize key tribological parameters using the Taguchi approach. The outcomes of this research contribute to the development of high-performance polyester powder coatings suitable for tribologically demanding industrial applications [16-22].

2. Materials and Experimental Methodology

2.1 Materials

A commercially available thermosetting polyester powder coating was used as the matrix material in the present study. Boron carbide (B_4C) nanoparticles were selected as the reinforcing nanofiller due to their extremely high hardness, low density, excellent wear resistance, and chemical

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inertness. The B_4C nanoparticles had an average particle size in the nanometer range and were used without further surface modification.

Mild steel (IS 2062 E250) plates were used as the substrate material owing to their widespread industrial application and good compatibility with powder coating processes. The chemical composition and mechanical properties of the substrate complied with relevant standards.

2.2 Substrate Preparation

Rectangular mild steel specimens of dimensions 20 mm × 20 mm × 5 mm were prepared for coating. Prior to coating, the substrates were mechanically polished using successive grades of emery paper to remove surface irregularities and oxide layers. The polished specimens were then degreased using acetone to eliminate oil and contaminants, followed by rinsing with distilled water and air drying.

Surface roughening was carried out to improve coating adhesion. The prepared substrates were stored in a dust-free environment until the coating process.

2.3 Preparation of B_4C-Reinforced Polyester Powder

The required quantity of B_4C nanoparticles was incorporated into the polyester powder using a hot-mixing technique. The polyester powder was heated to 72 ± 1 °C and mixed with B_4C nanoparticles at a constant rotational speed of 40 rpm for 15 minutes. This process facilitated uniform dispersion of nanoparticles within the polymer powder while avoiding premature curing or degradation.

Different weight percentages of B_4C reinforcement were prepared based on the experimental design. The prepared powder mixtures were cooled to room temperature and sieved to break agglomerates prior to coating.

2.4 Powder Coating Deposition and Curing

The B_4C-reinforced polyester powders were deposited on the prepared mild steel substrates using an electrostatic spray coating system operating under corona charging mode. The coated specimens were then cured in a hot air oven at a temperature of 180 °C for 15 minutes to ensure complete melting, flow, and crosslinking of the coating.

The final coating thickness was maintained at approximately 70 µm, measured using a coating thickness gauge. Uniform thickness was ensured for all samples to maintain consistency in tribological testing.

2.5 Experimental Design

Initially, a One-Variable-At-a-Time (OVAT) approach was adopted to determine the feasible range of process parameters. Based on OVAT results, three key factors were selected for optimization:

1. B_4C reinforcement content (wt.%)
2. Applied normal load (N)
3. Sliding frequency (Hz)

A Taguchi L9 orthogonal array was employed to systematically study the influence of these parameters with three levels each. This approach significantly reduced the number of experiments while ensuring reliable analysis of parameter effects.

2.6 Wear Testing

Wear behavior of the coated specimens was evaluated using a reciprocating tribometer under dry sliding conditions at room temperature. The tests were conducted against a hardened steel counterface. The wear rate was calculated based on volumetric material loss per unit load and sliding distance (mm^3/Nm).

The signal-to-noise (S/N) ratio was determined using the "smaller-is-better" criterion to identify optimal wear performance.

2.7 Scratch Hardness Testing

Scratch hardness was evaluated using a **scratch tester** by applying a progressively increasing normal load on the coated surface. The width of the scratch track was measured using optical microscopy, and the scratch hardness was calculated accordingly. The "larger-is-better" criterion was used for S/N ratio analysis of scratch hardness.

2.8 Statistical Analysis

Statistical analysis was carried out using Taguchi methodology and analysis of variance (ANOVA) to quantify the contribution of each parameter to wear resistance and scratch hardness. ANOVA was used to determine the significance of process parameters at a confidence level of 95%.

3. OVAT Analysis

3.1 Introduction

One variable at a time in this analysis a range of values are considered to identify the actual range of the variables were the required or optimized value can be found. The factors for the OVAT analysis are % weight of B4C nanoparticles, Load, Frequency and all other factors are kept constant

3.2 OVAT Analysis

In OVAT Analysis, each selected parameter is tested for 5 trials keeping the other parameters at minimum levels. The readings of each trial were taken to calculate Wear Rate

3.2.1 Selection of values

The basic criteria for selection of levels of factors for tribometer of various parameters is selected from technology guidelines of machine

Reinforcement: 2%, 4%, 6%, 8%, 10%

Load: 5, 10, 15. 20, 25N

Frequency: 1, 2, 3, 4, 5 Hz

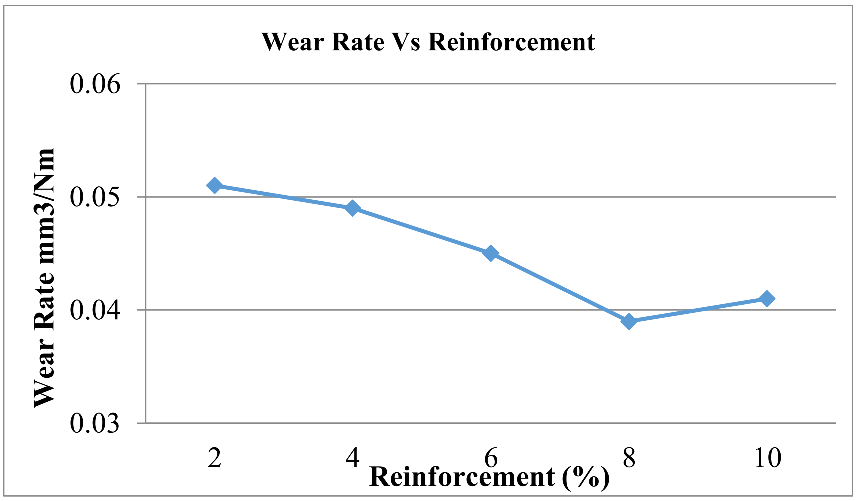
3.2.2 OVAT for Reinforcement

Table 1 *OVAT for Reinforcement*

Sr. No.	Reinforcement (%)	Wear Rate (mm^3/Nm)
1	2	0.051
2	4	0.049
3	6	0.045
4	8	0.039

5	10	0.041

Variation in Wear Rate with change in Reinforcement is shown in graph 1



Graph 1 *Wear Rate Vs Reinforcement*

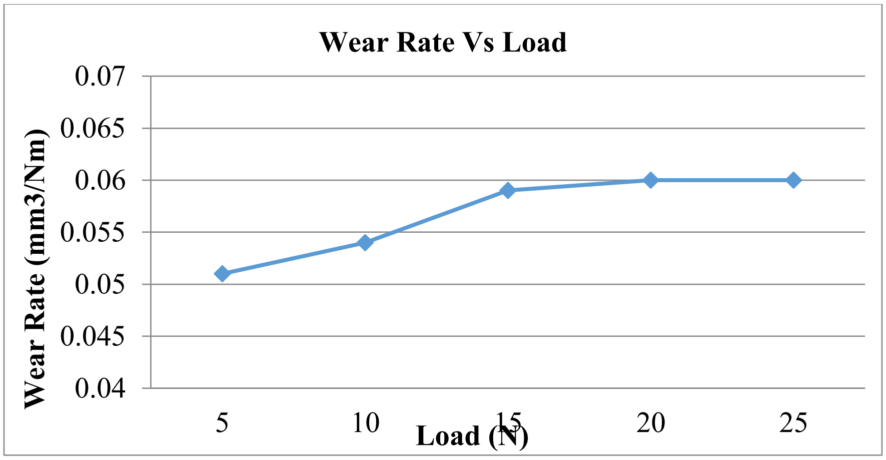
Load and Frequency kept constant and reinforcement varied from 2 to 10%. From the Graph 1, it has been observed that as reinforcement increases with decreasing wear rate, also has been observed that, most affected area and the rate of change of Wear Rate is higher in the region of 4 to 8% hence this level of factor has been selected.

3.2.3 OVAT for Load

Table 2 *OVAT for Load*

Sr. No.	Load (N)	Wear Rate (mm^3/Nm)
1	5	0.051
2	10	0.054
3	15	0.059
4	20	0.060
5	25	0.060

Variation in Wear Rate with change in Load is shown in graph 2.



Graph 2 *Wear Rate Vs Load*

Reinforcement and Frequency kept constant and Load varied from 5 to 25 N. From the graph 2, it has been observed that as Load increase from 5 to 25 N with increasing wear rate, also has been observed that, most affected area and the rate of change of Wear Rate is higher in the region of 5 to 15 N hence this level of factor has been selected.

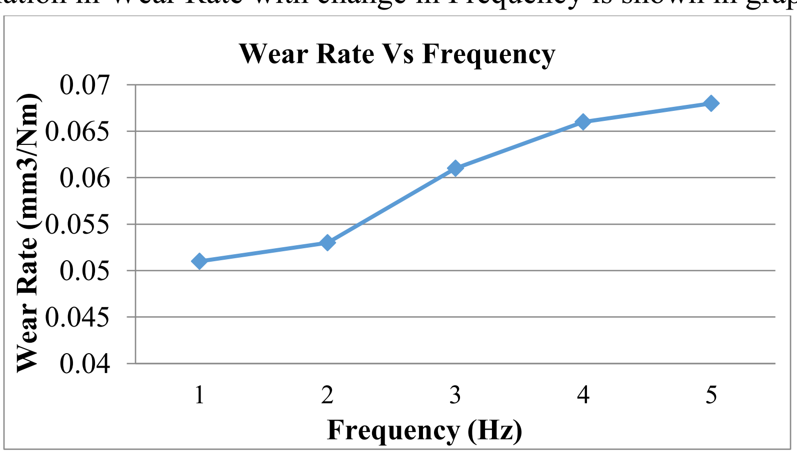
4.2.4 OVAT for Frequency

It is denoted by T and expressed in minute. Increase in the Frequency value will decrease the Wear Rate which in turn can improve the quality characteristics of buffs wheel further show in table 3.

Table 3 *OVAT for Frequency*

Sr. No	Frequency (Hz)	Wear Rate (%)
1	1	0.051
2	2	0.053
3	3	0.062
4	4	0.066
5	5	0.068

Variation in Wear Rate with change in Frequency is shown in graph 3.



Graph 3 *Wear Rate Vs Frequency*

Reinforcement and Load kept constant and Frequency varied from 1 to 5 Hz. From the graph 3, it has been observed that Frequency increases from 1 to 5 Hz with increasing wear rate also has been observed that, most affected area and the rate of change of Wear Rate is higher in the region of 2 to 4 Hz. hence this level selected.

4.3 Selections of Levels

Table 4 *Levels of input parameters*

Sr. No	Level 1	Level 2	Level 3
Reinforcement (%)	4	6	8
Load (N)	5	10	15
Frequency (Hz)	2	3	4

3.4 Standard L9 Orthogonal Array

While there are many standard orthogonal matrices available, each of the matrices is intended for a specific number of variables and independent design levels.

The standard notation for orthogonal arrays is

Ln (Xm)

Where, n = Number of experiments to be carried out

X = Number of levels

m = Number of factors

The common orthogonal matrices are listed below for a quick reference.

(Arrays of 2 levels) --- L4 (23), L8 (27), L12 (211), L16 (215), L32 (231), L64 (263) etc.

(Arrays of 3 levels) --- L9 (34), L18 (21 * 37), L27 (313), L54 (21 * 325), L81 (340) etc.

(Arrangements of 4 levels) --- L16 (45), L32 (21 * 49) etc.

Note: The matrices L18 (21 * 37), L54 (21 * 325), L32 (21 * 49) etc. They are for mixed level factors. In this investigation, carried out for 3 factors (Reinforcement, Load and Frequency each factor of 3 levels is chosen, an orthogonal L9 matrix (34) to perform the experiments.

Table 4.5 shows an orthogonal L9 matrix. A total of 9 experiments will be carried out and each experiment is based on the combination of the level values shown in the table. In this study, the Taguchi L9 OA is used with repetitions of the repeated reaction method to record the data of the experimental results. Repeated measurements are taken

Table 4 *Standards L9 orthogonal array*

Sr. No.	Reinforcement (%)	Load (N)	Frequency (Hz)
1	4	5	2
2	4	10	3
3	4	15	4
4	6	5	3
5	6	10	4
6	6	15	2
7	8	5	4

8	8	10	2
9	8	15	3

4. RESULTS AND DISCUSSION

4.1 Experimental results for Wear Rate

The Table 5 shows the L9 orthogonal array with repeat measurement of responses for runs one to nine. Repeats of response measurement technique is used overcome the drawback of saturated design in Minitab 19 software. It also shows that the S/N ratio for run one and ten are same as it is calculated for the repeats measurement. The S/N ratio values are calculated with help of Minitab19 software.

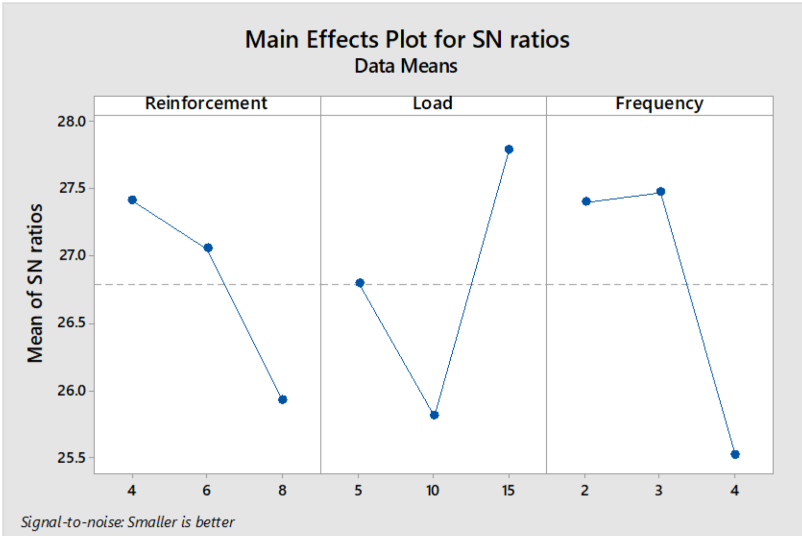
Table 5 *Cumulative data for S/N ratio Wear rate*

Experiments	**Inputs Factors**			**Output Responses**	
Trial No.	**Reinforcement (%)**	**Load (N)**	**Frequency (Hz)**	**Wear rate (mm^3/Nm)**	**S/N Ratio**
1	4	5	2	0.040	27.9588
2	4	10	3	0.043	27.3306
3	4	15	4	0.045	26.9357
4	6	5	3	0.042	27.5350
5	6	10	4	0.058	24.7314
6	6	15	2	0.037	28.8739
7	8	5	4	0.057	24.8825
8	8	10	2	0.054	25.3521
9	8	15	3	0.042	27.5350

4. 2 Main effects of Wear Rate

The influence of each control factor (Reinforcement, Load and Frequency) Wear Rate was analyzed from the S/N ratio response table, which expresses the S/N ratio at each level of control factor. From the graphs the optimized value is found where the wear rate is minimum and at what % of reinforcement the coating performs better. At 4 % weight Reinforcement, 15N load and at 3 Hz frequency the wear rate will low so conformation test will be performed to check the validity of the information from the graph

A higher S/N ratio with better characteristics was used and was calculated with the help of the Minitab19 software for the experimental tests. The values of the complete S/N ratio are S/N ratio for the first test. From the graph 4.4, it is observed that the optimum Wear Rate was in the highest values of the in the response graph. The configuration of the operating parameters with the highest ratio always provides the optimum quality with a minimum variation. The graph shows the relationship change when the control factor configuration was changed from one level to another. The main effect plots for mean of S/N ratio of Wear Rate is as shown in graph 4.4



Graph 4 *Main effect plots for mean of S/N ratio of Wear Rate*

4.8 Analysis of variance (ANOVA)

ANOVA, the ratio between the variance of cutting parameter and the error variances is called Fisher ratio (F). It is used to determine whether the parameter has significant effect on the quality characteristic from comparing the F test value of the parameter with the F standard table value at P significance level. If the F test value is greater than P test the operating parameter is considered significant. Relevance of the models is tested from ANOVA. It is a statistical tool for testing the null hypothesis for planned experiments, in which several different variables are studied simultaneously. ANOVA is used to quickly analyze the variances in the experiment using the Fisher test (F test). ANOVA table shown the result of the ANOVA analysis. ANOVA analysis makes it possible to observe that the value of p is less than 0.05 in the three parametric sources. It is therefore clear that Reinforcement, Load and Frequency of the material have an influence on the buffs wheel material. The last column of cumulative ANOVA has shown the percentage of each factor in the total variance that indicates the degree of impact on the outcome.

It shows table 4.7 that the Reinforcement (21.54%), the Load (33.79%) and the Frequency (43.87%) have major influence on the Wear Rate. Contribution of Frequency (43.87%) is highest among all three parameters hence it is most dominating parameter while reinforcements is least affecting parameter.

Table 6 *ANOVA results of Wear Rate*

Source	DF	Adj SS	Adj MS	F-Value	P-Value	% Contribution
Reinforcement	2	0.000109	0.000054	23.29	0.041	21.54
Load	2	0.000171	0.000085	36.57	0.027	33.79
Frequency	2	0.000222	0.000111	47.57	0.021	43.87
Error	2	0.000005	0.000002			
Total	8	0.000506				

4.9 Development of regression model for Wear Rate

Regression model has been developed using Minitab 19 software. Substituting the experimental values of the parameters in regression equation, values for Wear Rate have been predicted for all levels of study parameters. Graphical representation also shows that a predicted and experimental value of Wear Rate correlates with each other.

To establish the correlations between the parameters are (1) Reinforcement, (2) Load, (3) Frequency, of buff wheel material the model of multiple linear regressions was obtained using the statistical software "Minitab19". Terms that are statistically significant are included in the model. The final equation obtained is the following.

Mathematical models of Reinforcement, Load and Frequency is calculated using Minitab19 software and analysis of regression carried out to obtain the predicted value of Wear Rate.

Regression Equation

$$\text{Wear Rate} = 0.0242 + 0.00208\ \text{Reinforcement} - 0.000533\ \text{Load} + 0.00500\ \text{Frequency}$$

Table 6 gives comparison between experimentally measured and predicted Wear Rate by developed mathematical equation

***Table** 7 Experimental and predicted values of Wear Rate*

Sr. No.	Experimental value	Predicted value	Error %
1	0.040	0.039	2.54
2	0.043	0.042	3.38
3	0.045	0.044	2.27
4	0.045	0.049	8.16
5	0.056	0.051	9.80
6	0.037	0.038	5.26
7	0.057	0.058	1.72
8	0.049	0.045	8.88
9	0.047	0.052	9.61

Difference between Wear Rate values calculated using regression equation and experimental values for each experience found less than 10%. Hence, we can say that the regression equation developed is valid. Graph 5 shows the graphical representation of experimental and predicted values calculated using regression equation.

Taguchi design and ANOVA give mathematical model which predict the result nearly accurate. This can be due to this project deals the parameter level in very short range. Also, every effort has done to maintain noise factors constant to greater level throughout experimentation Wear Rate decreases for first sample and increases for ninth sample because of experiment from 1-9 are the set of experiments from Taguchi design L9 array; every experiment has different level of parameters and having unique combination of parameter level. The comparison between experimental and predicted value of Wear Rate is as shown in graph.

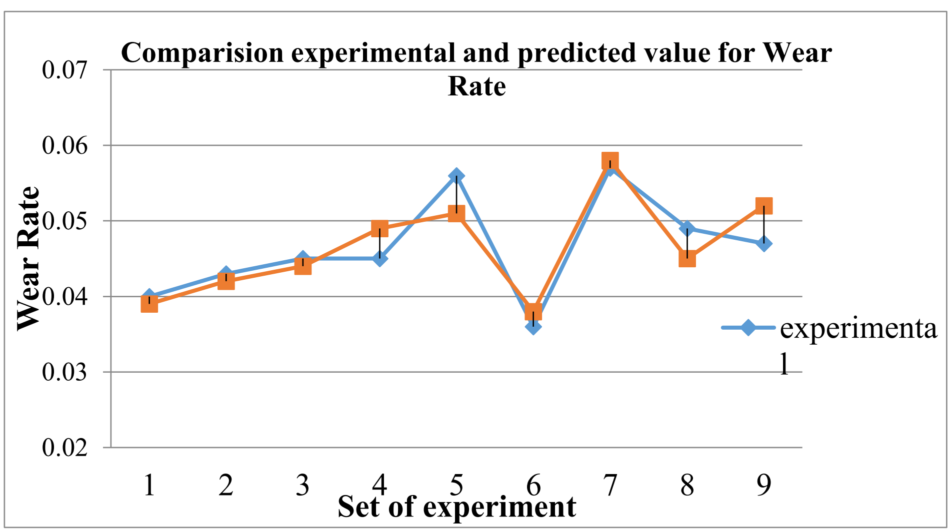
4.10 Confirmation experiment result for Wear Rate

Table 8 shows the difference between value of Wear Rate of confirmation experiment and value predicted from regression model developed.
Experiments was conducted for Reinforcement at level 3, Load at level 2 and Frequency at level

Table 8 *Confirmation experiment result for Wear Rate*

Parameter	Model value	Experimental value	Error %
Wear Rate (mm^3/Nm)	0.038	0.036	5.26

Confirmation experiment is conducted by keeping parameters at optimum levels suggested by Taguchi method and the Wear Rate value obtained has been compared with value predicted by the regression model keeping the parameters at same levels. It can be seen that the difference between experimental result and the predicted result is 5.26 %. This indicates that the experimental value correlates to the estimated value



Graph 4.14 *Comparison between experimental and predicted value of Wear Rate*

5. Conclusion

In this study the influence of operating parameters such as Reinforcement, Load, and Frequency and their optimization and

ANOVA tool were used to find the significant quantity of nanoparticles which can be added to the polyester powder coating to improve its wear resistance. Following conclusions are drawn.

1) From experimentation the optimum value was found to be 4 % weight reinforcement, 15 N load and 3 Hz frequency. But the most important result was that 4% weight of reinforcement shows better wear resistance than the polyester powder and 10% weight reinforcement powder. And ANOVA analysis for wear resistance shows the percentage contribution and for reinforcement it was 21.54 %.
2) ANOVA results indicate that frequency plays prominent role in determining the Wear Rate. The contribution of Reinforcement, Load and Frequency to the quality characteristics Wear Rate is 21.54%, 33.79% and 43.77% respectively.
3) There was a great improvement in the wear resistance of the 4 % weight of B_4C coating compared to the others coatings. The boron carbide reduces the contact area between the coating and the sliding pin which helps to delays the onset of wear during reciprocating abrasion test and also decreases the wear rate.
4) The addition of nanoparticles above 4 % hinders the electrostatic charging process of the powder. Due to which it affects the mechanical properties of the coating
5) Hot mixing of the powder with B_4C helped in homogenous mixing of nanoparticles. Mixing of nanoparticles does not affect the adhesiveness of the coating
6) Value of Wear Rate is lower obtained in confirmation experiment. Hence, good quality of polyester coating with B4C can be achieved using suggested level of parameters by Taguchi method.
7) Values of Wear Rate calculated using regression model correlates with experimental values with error less than 10%. Hence the model developed is valid and experimental results of Wear Rate with any combination of operating parameters can be estimated within selected levels.

6. Conflict of Interest

The authors declare that there is no conflict of interest regarding the publication of this research.

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